

Relay Retry Semantics and Dead Letter Handling

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When a delivery attempt fails, Relay retries with exponential backoff. There are six retry attempts following the initial delivery, so seven attempts in total, spread over a little under two hours. Each interval carries up to 20 percent random jitter, which exists to prevent synchronised retry waves across tenants after a shared dependency recovers; without jitter, every endpoint that failed during the same window would retry in lockstep and reproduce the original overload.

Attempt	Delay after previous attempt	Approximate elapsed time
1 (initial)	—	0
2	2 seconds	2 seconds
3	8 seconds	10 seconds
4	30 seconds	40 seconds
5	2 minutes	2 minutes 40 seconds
6	15 minutes	17 minutes 40 seconds
7 (final)	1 hour	1 hour 17 minutes

Table 1. Retry schedule. Intervals carry up to 20 percent jitter, so elapsed times are approximate.

A failure is any non-2xx response, any connection error, or any response taking longer than 10 seconds. Relay does not distinguish between a 500 and a 404 for retry purposes, and this is intentional rather than an oversight: an endpoint returning 404 during a deploy is extremely common, and treating a 404 as permanent would silently drop legitimate events during every deployment a customer performs. Customers occasionally request configurable per-status retry behaviour and it has been declined three times.

Dead letter queue

After the final attempt fails, the event moves to the dead letter queue for that endpoint. Dead lettered events are retained for 14 days on all plans, including Starter, and this retention is not affected by the plan's event history retention, which is a separate and longer-lived record. Dead lettered events can be replayed individually or in bulk from the dashboard or through the API. Replaying resets the attempt counter, so a replayed event that fails again proceeds through the full seven-attempt sequence a second time and, if it fails again, returns to the dead letter queue with its retention clock restarted.

Circuit breaker

Each endpoint has its own circuit breaker. Since 19 February 2026 it opens when the failure rate exceeds 50 percent over a rolling 60 second window, subject to a minimum of 20 attempts within that window to avoid opening on low-volume noise. Prior to that change it required 20 consecutive failures, a threshold that failed to catch endpoints which were failing at high rates while intermittently succeeding, and which was the direct cause of the February 2026 regional backlog incident.

While the breaker is open, deliveries to that endpoint are held rather than attempted, and a single probe request is sent every 30 seconds until one succeeds. Held events are not consuming retry attempts and do not advance toward the dead letter queue while the breaker is open; the retry clock resumes when the breaker closes. The period during which held events are delivered after the breaker closes is referred to internally as the flush window, and it is frequently mistaken for a rate limit by customers watching their own inbound traffic.

Tenant isolation

Since 19 February 2026 delivery queues are partitioned per tenant and scheduled fair-share. One tenant's retry volume can no longer delay deliveries belonging to other tenants in the same region, which was possible under the previous shared-queue architecture and is what turned a single customer's endpoint failure into a regional incident. In addition, a tenant whose retry volume exceeds ten times its own first-attempt volume is now automatically rate limited, and the account owner is notified at the point the limit engages rather than afterwards.

Ordering

Ordering is preserved per endpoint but not across endpoints. If event A fails and event B succeeds, B is not delivered ahead of A to the same endpoint: subsequent events queue behind the retrying event until it either succeeds or is dead lettered. Two events published simultaneously to two different endpoints have no ordering relationship at all. This surprises people, and it is by a wide margin the most common support question about retries, ahead of dead letter retention and ahead of the circuit breaker. It is also the property most worth explaining carefully rather than linking to, because the consequence of misunderstanding it is a receiver built on an assumption that holds until the first failure and then stops holding.